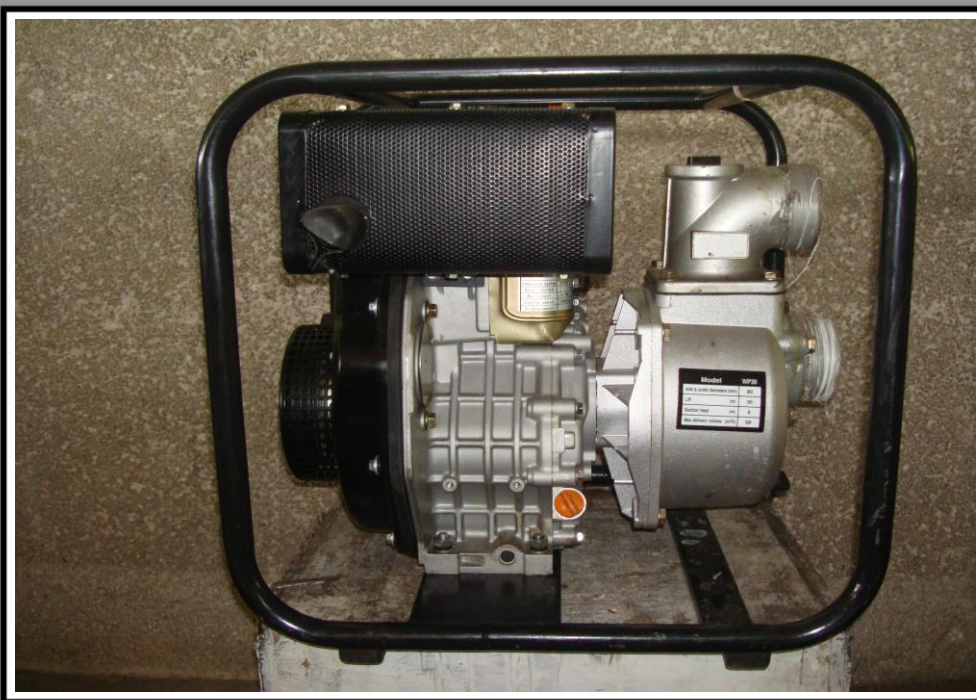




AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 - 33



OKAMA CENTRIFUGAL PUMPSET Model : KDP3

OKAMA KDP3 Self-priming Centrifugal Pumpset

Test Report No.	:	2013 - 33
Date of Test	:	March 18, 2013
Test Valid Until	:	March 18, 2018
Brand	:	OKAMA
Model	:	KDP3
Serial No.	:	No data
Type and Size	:	Single-end suction, 75 x 75 mm, Self-Priming Centrifugal pumpset
Manufacturer	:	FUZHOU LEADING POWER CO., LTD. China
Test Requested by	:	SERROTMA TRADING ENTERPRISES, INC. 2123H Juan Luna St., Tondo, Manila
Machine Classification	:	Commercial

SUMMARY

The OKAMA KDP3 Self-priming centrifugal pumpset was tested for performance and suction condition. It was tested at the recommended full throttle speed setting of the engine. The engine and pump were mounted on a steel frame.

The Maximum Total Head (H) was 24.1 m at zero discharge condition. The shaft speed of the pumpset at this point was 3682 rpm, with Engine Fuel Consumption Rate of 1.07 L/h. The Maximum Discharge Capacity of 17.8 L/s occurred at 6.40 m Total Head with Fuel Consumption Rate of 1.22 L/h. The shaft speed at this point was 3566 rpm. The Maximum Fuel Consumption Rate was 1.40 L/h occurred at 16.0 L/h Discharge Capacity and 10.8 m Total Head. The Maximum System Efficiency was 13.6% occurred at 11.8 L/h Discharge Capacity and 16.3 m Total Head. The Engine Shaft Speed at this point was 3638 rpm.

The suction condition of the pump was also tested. With zero discharge, the Maximum Total Suction Lift was 7.48 m. The Net Positive Suction Head Available at this point was 2.40 m.

PURPOSE AND SCOPE OF TEST

The purpose of the test was to establish the pumpset performance which included the Fuel Consumption Rate and the Total Head at varying discharge capacities and suction conditions. The pumpset was mounted on the laboratory test set-up shown in **Figure 1**.

The test was conducted on March 18, 2012 at the AMTEC Test Laboratory, UPLB, College, Laguna.



Figure 1. OKAMA KDP3 Pumpset on the test set-up

METHOD OF TEST

The pump was tested following the “ Philippine Agricultural Engineering Standards, PAES 115:2000, Agricultural Machinery – Centrifugal, Mixed-Flow, Axial-Flow Water Pumps – Methods of Test”.

DESCRIPTION OF THE PUMP

The OKAMA KDP3 Self-priming Centrifugal pumpset shown in **Figure 2**, was a single-end suction, 75 x 75 mm, centrifugal pump close-coupled to a 4.40 kW (5.90 hp) OKAMA diesel engine, Model 178F. The pumpset was mounted on a steel base. The suction inlet and discharge outlet were threaded flanges and designed for a 75 mm diameter pipe. The impeller was a semi-open type made of cast iron material connected to the pump shaft through a thread. It was housed in a cast iron double volute casing enclosed in an aluminum chamber. The priming inlet was located on top of the pump beside the discharge outlet. The suction inlet was provided with a swing type one-way valve to retain the sufficient amount of liquid inside the chamber necessary to prime the pump. The discharge outlet was located on top of the pump facing upward. The stuffing box used a mechanical seal.

The primemover, an OKAMA Diesel engine, Model: 178F was a four-stroke, air-cooled, single vertical cylinder diesel engine. Fuel feed system was of gravity type with fuel tank located on top of the engine. It utilized a dry type air cleaner using pleated paper element. Lubrication system was force-feed lubricating type. Cooling was done by air forced by a fan/flywheel encased in a metal shroud. It was started using a rope-recoil system. A muffler was used in the exhaust system.

Table 1 shows the detailed specifications of the pump.



Figure 2. OKAMA KDP3 Self-priming Centrifugal Pumpset

PUMPSET SPECIFICATIONS

Table 1. Detailed specifications of the OKAMA KDP3 Self-priming Centrifugal Pumpset

1.	Brand	:	OKAMA
2.	Model	:	KDP3
3.	Serial No.	:	No data
4.	Make	:	FUZHOU LEADING POWER CO., LTD. China
5.	Type	:	Single-end suction, Self- priming centrifugal pumpset
6.	Size (mm)	:	75 x 75
7.	Overall Dimension and Weight (Including mounting frame and engine)		
7.1	Length (mm)	:	537
7.2	Width (mm)	:	460
7.3	Height (mm)	:	500
7.4	Weight (kg)	:	56.0
8.	Impeller		
8.1	Type	:	Semi-open
8.2	Diameter (mm)	:	122
8.3	Width (mm)	:	24
8.4	No. of vanes	:	4
8.5	Material	:	Cast iron
9.	Suction Inlet		
9.1	Type	:	Threaded flange
9.2	Diameter (mm)	:	75
9.3	Material	:	Aluminum
10.	Discharge Outlet		
10.1	Type	:	Threaded flange
10.2	Diameter (mm)	:	75
10.3	Material	:	Aluminum

Table 1. (Cont'n)

11.	Casing		
11.1	Type	:	Double volute
11.2	Material	:	Cast Iron
12	Stuffing box	:	Mechanical seal
13.	Rotation	:	Counter clockwise when facing the suction inlet
14.	Rated speed (rpm)	:	3600
15.	Rated Maximum Discharge Capacity (L/s)	:	16.1
16.	Rated Maximum Total Head (m)	:	30.0
17.	Primemover (Based on nameplate)		
17.1	Brand	:	OKAMA
17.2	Model	:	178F
17.3	Serial No.	:	no data
17.4	Make	:	FUZHOU LEADING POWER CO., LTD. China
17.5	Type	:	Four stroke, air-cooled single vertical cylinder diesel engine
17.6	Rated Output Power (kW)		
	Maximum	:	4.40 @ 3600
	Continuous	:	4.0 @ 3600 rpm
17.8	Bore x Stroke	:	78 x 62 mm
17.9	Displacement Volume (cc)	:	296
17.10	Cooling System	:	Air-cooled
17.11	Lubricating System	:	Force-feed type
17.12	Starting System	:	Rope-recoil
17.13	Air Cleaner	:	Dry type (cylindrical pleated paper element)
17.14	Exhaust System	:	Muffler

PERFORMANCE TEST

Results of performance test are shown in **Table 2** and plotted in **Figure 3**.

Pumpset Tested : OKAMA KDP3

Date of Test : March 18, 2013

Test Condition

Ambient Temperature (°C) : 33.3

Atmospheric Pressure (mb) : 1013

Pumping Water Temperature (°C) : 35.5

Table 2. Results of Performance Test

PUMP SHAFT SPEED (rpm)	FUEL CONSUMP- TION (L/h)	TOTAL HEAD (m)	DISCHARGE CAPACITY (L/s)	WATER POWER (kW)	INPUT POWER (kW)	SYSTEM EFFICIENCY *%)
3682	1.07	24.1	0	0	11.1	0
3668	1.08	22.0	3.55	0.77	11.2	6.85
3633	1.18	20.0	7.23	1.42	12.3	11.6
3641	1.27	18.1	6.38	1.13	13.2	12.9
3638	1.34	16.3	11.8	1.89	13.9	13.6
3631	1.37	14.4	13.2	1.88	14.2	13.2
3622	1.39	12.6	14.7	1.81	14.5	12.5
3619	1.40	10.8	16.0	1.68	14.6	11.6
3601	1.38	8.90	16.9	1.48	14.5	10.3
3566	1.22	6.40	17.8	1.12	12.7	8.81

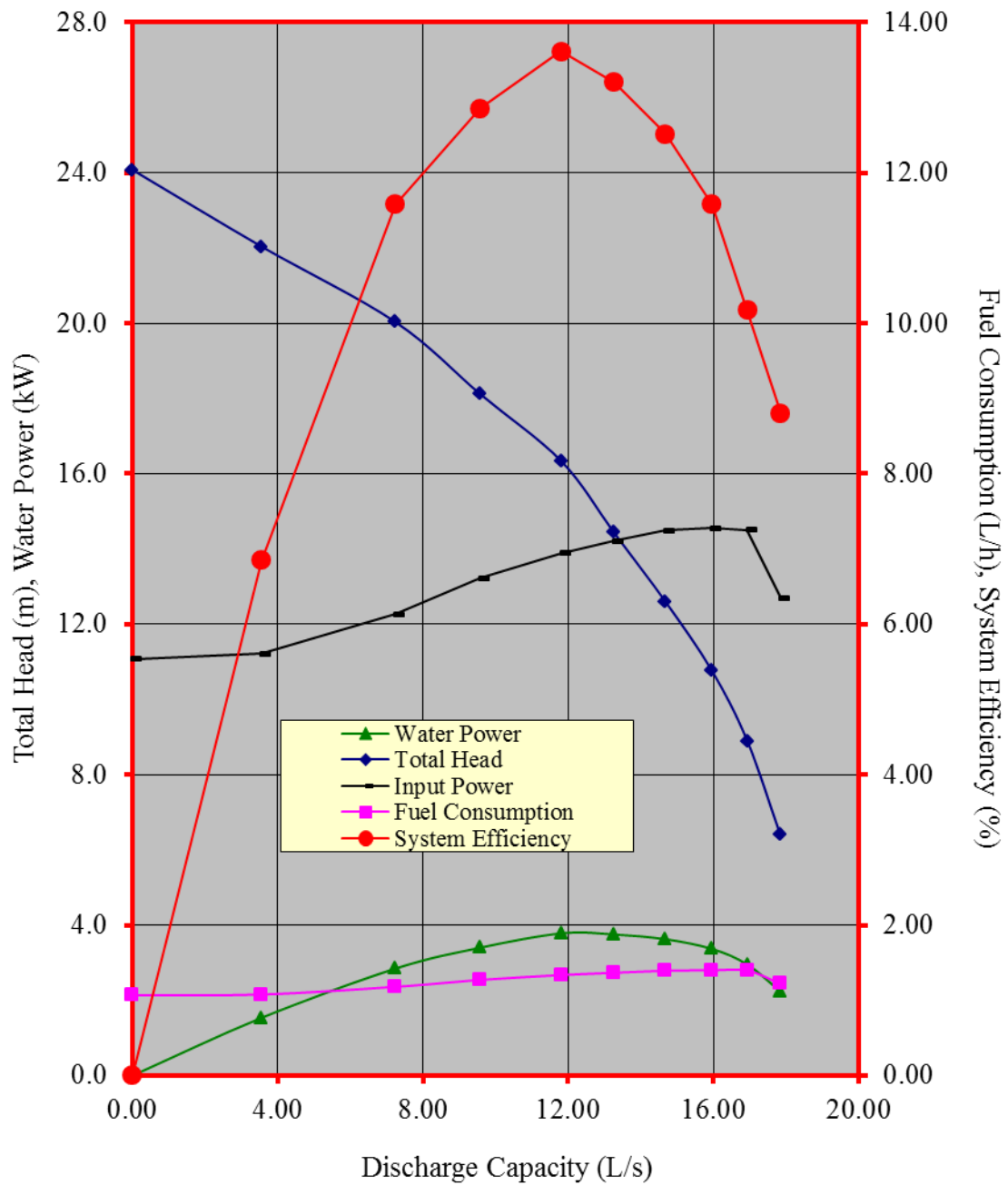


Figure 3. Performance characteristic curves of the OKAMA KDP3 self-priming centrifugal pumpset

CAVITATION TEST

Results of Cavitation Test are shown in **Table 3**.

Pumpset Tested : OKAMA KDP3

Date of Test : March 18, 2013

Test Condition

Ambient Temperature (°C) : 33.3

Atmospheric Pressure (mb) : 1013

Pumping Water Temperature (°C) : 35.8

Table 3. Results of Cavitation Test

PUMP SHAFT SPEED (rpm)	TOTAL SUCTION LIFT (m)	NET POSITIVE SUCTION HEAD AVAILABLE (m)	TOTAL HEAD (m)	DISCHARGE CAPACITY (L/s)
3682	1.31	8.56	36.1	17.9
3668	2.73	7.14	34.3	16.4
3633	4.37	5.50	32.7	12.6
3641	5.95	3.92	31.1	8.29
3638	6.69	3.18	28.8	6.45
3631	7.48	2.40	26.4	0

DATA COMPUTATION

ANNEX 1. Formula used during calculation and testing

1. Total Head (m), H

$$TDH = H_d - H_s$$

$$\text{Where: } H_d = \frac{P_2}{\gamma} + \frac{V_2^2}{2g} + Z_2$$

$$H_s = \frac{P_1}{\gamma} + \frac{V_1^2}{2g} + Z_1$$

H_d = Total Discharge Head (m)

H_s = Total Suction Lift/Head (m)

$\frac{P_2}{\gamma}$ = Pressure gage reading on the discharge pipe at
gage connection converted to meter

$\frac{P_1}{\gamma}$ = Vacuum gage reading on the suction pipe at the
point of gage connection converted to meter

$\frac{V_2^2}{2g}$ = Velocity head on the discharge pipe converted
to meter. Computed by determining the velocity
of liquid on the pipe by formula, $V = Q/A$ where
 V , velocity of liquid inside the pipe in meter per
per second, Q , Discharge rate of the pump in
liters per second, and A , Cross-sectional area
of the pipe in square meter.

$\frac{V_1^2}{2g}$ = Velocity head on the suction pipe converted to
meter

Annex (Cont'n)

Z_2 = The distance from the center of the pressure gage face to the centerline of the pump in meters

Z_1 = The distance from the center of the vacuum gage face to the centerline of the pump in meters

2. Discharge Capacity, Q , in liters per second (L/s) is determined through gravimetric method using a platform scale and a stopwatch.

3. Water Power, WP , is the output power of the pump in kilowatt (kW) and computed using:

$$WP = \frac{H \times Q}{102}$$

4. Input Power, IP , is the power on the input shaft of the pump. This is the power in kilowatt required to drive the pump and is determined by the formula:

$$IP = \frac{T \times N}{974}$$

Where:

T = Torque in kilogram-meter, on the input shaft of the pump and determined by a dynamometer

N = Input shaft angular speed in RPM

5. Pump Efficiency, η ,

$$\eta = \frac{WP}{IP}$$

6. Net Positive Suction Head Available, NPSHA. This is the net force (converted to meter) needed to push the water inside the pipe and computed by the formula:

$$NPSHA = P_{atm} - P_{vp} - H_s$$

Where:

P_{atm} = Atmospheric pressure during the test converted to meters

P_{vp} = Vapor pressure of the test liquid at certain temperature converted to meters

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Checked by:

DARWIN C. ARANGUREN

Approved for release:

DELFIN C. SUMINISTRADO
Director

Date

N.B.

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